TechLIZA Project Report

Fine Tuning a Small Language Model for Technology Interviews

Executive Summary

TechLIZA is a technology media interviewer built from Qwen2.5-0.5B-Instruct and adapted with LoRA supervised fine-tuning. I trained the model on follow-up question examples extracted from NPR technology interviews, compared one-turn and two-turn context designs, and tested whether the learned behavior transferred to interviews from another source. The final system uses the one-turn adapter because it used more training data and performed about as well as the two-turn model in blinded human evaluation.

Main result The fine-tuned model became more likely to ask about a specific detail in the interviewee's answer. In the external evaluation, TechLIZA was preferred over the base model in 7 of 10 cases. This is encouraging but preliminary because the evaluation sets were small and the judgments came from a limited manual review.

Project Objective

The project asked whether a small language model could learn useful interviewer behavior from real media interviews. The target behavior was narrow: given the previous interviewer question and the interviewee's latest answer, the model should ask exactly one concise, relevant follow-up question. This framing made it possible to compare the original model with a task-specific version while keeping the model size and prompt format constant.

Data Preparation and Experimental Process

I began with the NPR interview corpus, which contained 22,149 episodes. A broad keyword search returned 1,440 possible technology episodes, but terms such as internet, computer, and cyber also retrieved unrelated stories. I therefore applied higher-precision title patterns and manually inspected the results. This reduced the pool to 238 high-precision candidates and then to 220 technology-focused episodes.

Within each episode, consecutive turns from the same speaker were merged. I then extracted host to guest to host sequences and represented each example as the previous interviewer question plus the interviewee answer, with the next interviewer question as the target. The first extraction produced 961 candidates. Length limits and a requirement that the target contain exactly one question reduced the set to 551 examples, or 57.3 percent of the candidates.

Table 1 Dataset Construction and Split

Stage	Episodes	Examples	Purpose
Full NPR corpus	22,149	Not applicable	Starting corpus
Broad topic filter	1,440	Not yet extracted	Maximize recall
High precision filter	238	Not yet extracted	Remove obvious false positives
Refined technology set	220	961 candidates	Create interview triples

Stage	Episodes	Examples	Purpose
Clean final set	189 represented	551	Train and evaluate
Train split	151	436	LoRA training
Validation split	19	56	Loss monitoring
Test split	19	59	Held-out comparison

The split was performed at the episode level rather than the example level. This prevented different turns from the same interview from appearing in both training and evaluation data.

Model Training and System Development

The base model was Qwen2.5-0.5B-Instruct. LoRA adapters were added to the query, key, value, and output attention projections with rank 16, alpha 32, and dropout 0.05. Only 2,162,688 of 496,195,456 parameters were trainable, or 0.4359 percent, which made training practical on a Google Colab T4 GPU.

Training ran for three epochs with a learning rate of 2e-4, FP16 precision, a per-device batch size of 4, gradient accumulation of 4, and a maximum sequence length of 512 tokens. The first run used 436 training examples and 56 validation examples. The training loss fell from 2.8341 to 2.6181, while validation loss stayed nearly flat and ended slightly higher. The model learned the training distribution, but the loss curve did not show a clear improvement in held-out likelihood.

Table 2 First LoRA Training Run

Epoch	Training Loss	Validation Loss	Mean Token Accuracy
1	2.8341	2.9484	0.4091
2	2.6756	2.9485	0.4120
3	2.6181	2.9534	0.4096

Output inspection was more informative than validation loss alone. TechLIZA more often focused on a concrete term or claim, while the base model more often produced a broad question. The improvement was inconsistent: TechLIZA sometimes overused clarification prompts, repeated earlier material, or introduced a weak new direction.

Controlled Context Experiment

I next tested whether one additional question-answer turn would improve follow-up quality. Only 251 of 551 cleaned examples had valid two-turn context, a coverage rate of 45.6 percent. To isolate context length, both models used the same 198 training, 28 validation, and 25 test examples, as well as the same targets, LoRA settings, and random seed.

Table 3 Matched Context Experiment at Epoch 3

Model	Training Examples	Validation Loss	Human Preference
One-turn control	198	2.9231	11 of 25
Two-turn context	198	2.8502	10 of 25

Model	Training Examples	Validation Loss	Human Preference
Tie	Not applicable	Not applicable	1 of 25
Neither	Not applicable	Not applicable	3 of 25

The two-turn model's final validation loss was 2.5 percent lower than the one-turn control, but the blinded comparison did not show a meaningful qualitative advantage. The one-turn model won 44 percent of cases and the two-turn model won 40 percent. Because the original one-turn run had 436 training examples, more than twice the 198 examples available for the matched context experiment, I selected the original one-turn adapter for the final system.

Cross Source Evaluation

To test transfer beyond NPR, I compared the base model and final TechLIZA adapter on 10 held-out examples from the Anthropic Interviewer dataset. The cases covered creative work, scientific work, and the general workforce. Model identities were hidden during the manual A/B judgment. TechLIZA was preferred in 7 cases and the base model in 3.

Table 4 Blinded Human Evaluation Results

Evaluation	Sample Size	Result	Interpretation
Matched NPR context test	25	One-turn 11, two-turn 10, tie 1, neither 3	No clear benefit from adding a second turn
Anthropic cross-source test	10	TechLIZA 7, base model 3	Preliminary evidence of transfer beyond NPR

The cross-source result supports the main hypothesis, but it should not be treated as a precise estimate of performance. Ten examples are too few for a stable percentage, and manual preference reflects overall judgment rather than separate scores for relevance, specificity, depth, naturalness, and non-redundancy.

Final Application

The selected adapter was integrated into a Streamlit chat interface. The application loads the public Qwen base model, attaches the submitted LoRA adapter, and asks one follow-up question after each user response. The demonstration runs in Google Colab and exposes the local Streamlit server through a temporary Cloudflare URL. A relative adapter path was used so the submitted notebook, application file, and adapter folder can run together without relying on a personal Google Drive location.

Key Difficulties and Solutions

Table 5 Project Challenges and Responses

Difficulty	Why It Mattered	How It Was Addressed
Noisy topic filtering	Broad keywords retrieved many non-technology stories.	Used word-boundary title patterns, reviewed keyword groups, and refined 1,440 broad candidates to 220 episodes.
Irregular transcript structure	NPR data did not directly provide training triples.	Grouped by episode, restored turn order, merged consecutive same-speaker utterances, and extracted host-guest-host sequences.

Difficulty	Why It Mattered	How It Was Addressed
Data leakage risk	Adjacent turns from one interview could inflate evaluation results.	Split by episode before creating the train, validation, and test subsets.
Uneven example quality	Some answers were trivial and some targets contained multiple questions or long monologues.	Applied explicit word-count limits and required exactly one question in the target, retaining 551 of 961 candidates.
Library compatibility	The Colab environment included changing TRL interfaces and an incompatible optional torchao package.	Inspected the installed SFTConfig signature, recorded package versions, and removed the unused conflicting package.
GPU memory limits	Several base-model and adapter copies were needed for fair comparisons.	Used LoRA, FP16, short sequences, gradient accumulation, sequential model loading, and explicit CUDA cache cleanup.
Loss did not equal interview quality	The two-turn model had lower validation loss without a clear human preference advantage.	Added blinded A/B evaluation and an external dataset instead of selecting the model by loss alone.

Limitations

The training set was small and came from one media source, so it carried NPR's topic mix and interviewing style. The blind evaluations contained 25 and 10 examples and did not use multiple independent raters. The external result therefore shows promise rather than statistical proof. The model also retained visible failure modes, including generic questions, repetition, excessive clarification, and occasional topic drift. Finally, the temporary Cloudflare tunnel is appropriate for a classroom demonstration but not for persistent deployment.

A stronger follow-up study would expand the training corpus, use a fixed scoring rubric with several raters, report agreement between raters, and evaluate a larger cross-source sample. It would also test whether targeted data balancing can reduce repeated clarification patterns without losing the model's improved specificity.

Conclusion

TechLIZA showed that parameter-efficient fine-tuning can noticeably change the interviewing behavior of a 0.5-billion-parameter model. The final adapter trained only 0.44 percent of the model parameters, yet it produced more detail-focused follow-up questions and was preferred over the base model in 7 of 10 external cases. The context experiment also provided an important negative result: lower validation loss did not translate into a clear human preference for the two-turn design.

For that reason, the final design favors the original one-turn model. It uses a larger training set, requires less context, and achieved comparable human-rated quality. The project met its practical goal by combining a reproducible training pipeline, controlled comparisons, cross-source testing, and a working interactive demo. Its results support further study with larger datasets and more rigorous evaluation rather than claiming that the current model is complete.